

# Monte Carlo Results for the Two-Dimensional Random-Bond Ising Model

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## 1 Model and the Griffiths region

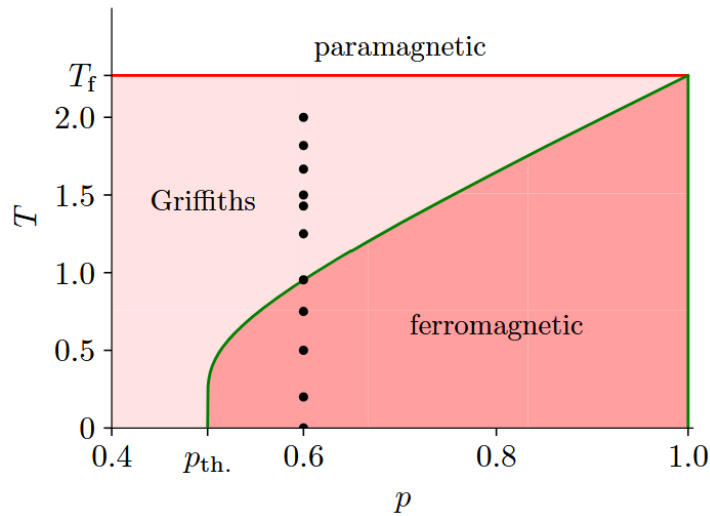
We consider spins  $s_i = \pm 1$  on an  $L \times L$  square lattice with periodic boundary conditions and  $N = L^2$  sites. Setting  $k_B = 1$ , the zero-field Hamiltonian is

$$H = - \sum_{\langle ij \rangle} J_{ij} s_i s_j, \quad J_{ij} = \begin{cases} 1, & \text{with probability } p, \\ 0, & \text{with probability } 1 - p. \end{cases} \quad (1)$$

Bonds are drawn independently and remain fixed during thermal sampling. This is the bond-diluted limit of the ferromagnetic random-bond Ising model. Dataset `run_004` uses  $p = 0.5$ . We denote thermal averages in realization  $a$  by  $\langle \cdot \rangle_a$  and disorder averages by  $[\cdot]_{\text{dis}}$ .

Disorder allows unusually well-connected, nearly pure regions inside a globally paramagnetic system. In the Griffiths region,  $T_c(p) < T < T_G$ , these rare regions can generate enhanced responses and slow relaxation, while the equilibrium free energy is nonanalytic in magnetic field at zero field. Here  $T_G = 2/\ln(1 + \sqrt{2}) \simeq 2.269$  is the pure square-lattice transition temperature. Classical Griffiths singularities can be weak: slow dynamics or broad response distributions alone do not establish their asymptotic form.

The present concentration equals the square-lattice bond-percolation threshold,  $p_c = 1/2$ , where the thermodynamic ferromagnetic transition temperature is zero. A heat-capacity maximum near  $T = 1$  therefore does not imply finite-temperature ordering. Critical connectivity, rare dense regions, and finite observation times must all be considered in interpreting the data.



**Figure 1:** Schematic temperature–bond-concentration phase diagram. The green curve denotes  $T_c(p)$ , separating ferromagnetic order from the paramagnetic phase. The red line is the pure-system transition temperature, labeled  $T_f$  in the image and denoted  $T_G$  here. The lightly shaded region between these boundaries is the Griffiths region. The label  $p_{\text{th.}}$  corresponds to the percolation threshold  $p_c = 1/2$ . Black markers illustrate a temperature scan at  $p = 0.6$ ; they are not the  $p = 0.5$  data reported below.

Figure 1 emphasizes that the Griffiths region is globally paramagnetic despite its slowly relaxing, locally correlated rare regions. At fixed  $p > p_c$ , cooling crosses  $T_G$  before reaching the ordering temperature  $T_c(p)$ . For the present  $p = p_c$ , the latter boundary reaches zero, leaving  $0 < T < T_G$  without bulk ferromagnetic order. The green curve is schematic and is not a fit to this dataset.

## 2 Monte Carlo procedure

The simulations used SpinMonteCarlo.jl 1.2.2. Each realization was thermalized with 2048 Swendsen–Wang (SW) updates: parallel neighboring spins are linked with probability  $1 - e^{-2J_{ij}/T}$ , and each cluster is independently assigned either orientation. Measurements then used 65536 random-site heat-bath sweeps. Each sweep selects  $N$  sites with replacement and resamples them according to

$$P(s_i = +1 \mid \{s_j\}) = \frac{1 + \tanh(h_i/T)}{2}, \quad h_i = \sum_{j:\langle ij \rangle} J_{ij}s_j. \quad (2)$$

Local responses also used SW thermalization followed by heat-bath measurements. This is the implementation that generated this dataset, before the subsequent change to SW sampling of static observables.

The sizes are  $L = 10, 16, 24, 32, 40, 50$ , with  $T = 0.01, 0.02, \dots, 3.00$  and 1000 disorder realizations per size. The same bond realizations are reused across temperatures. Measurements are divided into 256 blocks of 256 sweeps for block statistics and jackknife estimates. Disorder-average bands use the sample standard deviation divided by  $\sqrt{1000}$ ; this scatter already includes MC noise. A fixed block length need not resolve low-temperature slow modes and does not guarantee converged errors.

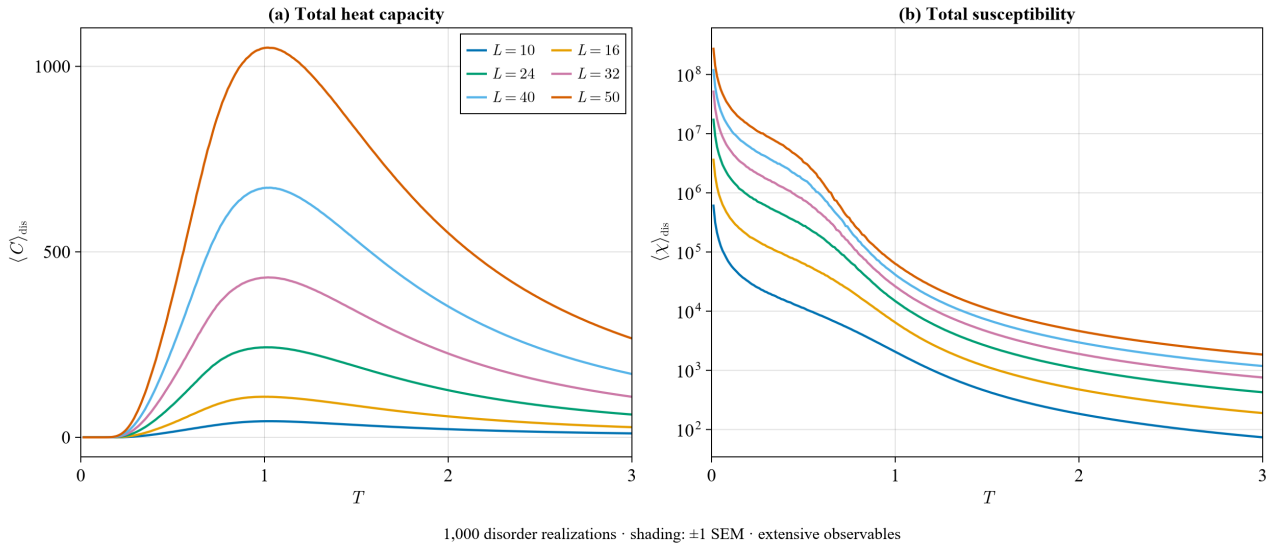
### 3 Results

#### 3.1 Disorder-averaged heat capacity and susceptibility

For  $M = \sum_i s_i$ , the observables are

$$C_a = \frac{\langle H^2 \rangle_a - \langle H \rangle_a^2}{T^2}, \quad \chi_a = \frac{\langle M^2 \rangle_a}{T}. \quad (3)$$

Both are total-system quantities; division by  $N$  gives per-site values. The susceptibility uses the exact symmetry  $\langle M \rangle_a = 0$  of the finite-volume zero-field ensemble, without subtracting  $\langle |M| \rangle_a^2$ .



**Figure 2:** Disorder-averaged total heat capacity (left) and total susceptibility (right) for six sizes. Shading denotes  $\pm 1$  standard error of the disorder mean (SEM). The susceptibility axis is logarithmic.

The heat capacity has a broad maximum at  $T = 0.99$ – $1.03$ , with little systematic peak displacement over these sizes. For  $L = 50$ , the maximum occurs at  $T = 1.02$ , where  $[C]_{\text{dis}}/N \simeq 0.420$ . The per-site peak decreases from approximately 0.435 at  $L = 10$  to 0.420 at  $L = 50$ . The growing total peak therefore mainly reflects increasing volume, rather than a growing per-site singularity. Its broad shape is consistent with a crossover in local bond-energy fluctuations. At low temperature, occupied bonds are mostly satisfied and energy fluctuations are suppressed; at high temperature the factor  $T^{-2}$  suppresses the heat capacity.

The susceptibility rises strongly upon cooling, without an isolated maximum in the displayed range. A useful low-temperature equilibrium picture is a set of nearly rigid disconnected clusters. A cluster of  $n_c$  spins contributes approximately  $n_c^2$  to  $\langle M^2 \rangle$ , giving  $\chi \simeq T^{-1} \sum_c n_c^2$  when cluster orientations are independently sampled. Large clusters have disproportionate weight, making magnetic response particularly sensitive to connectivity. This is a physical interpretation, rather than a quantitative fit to the trajectories.

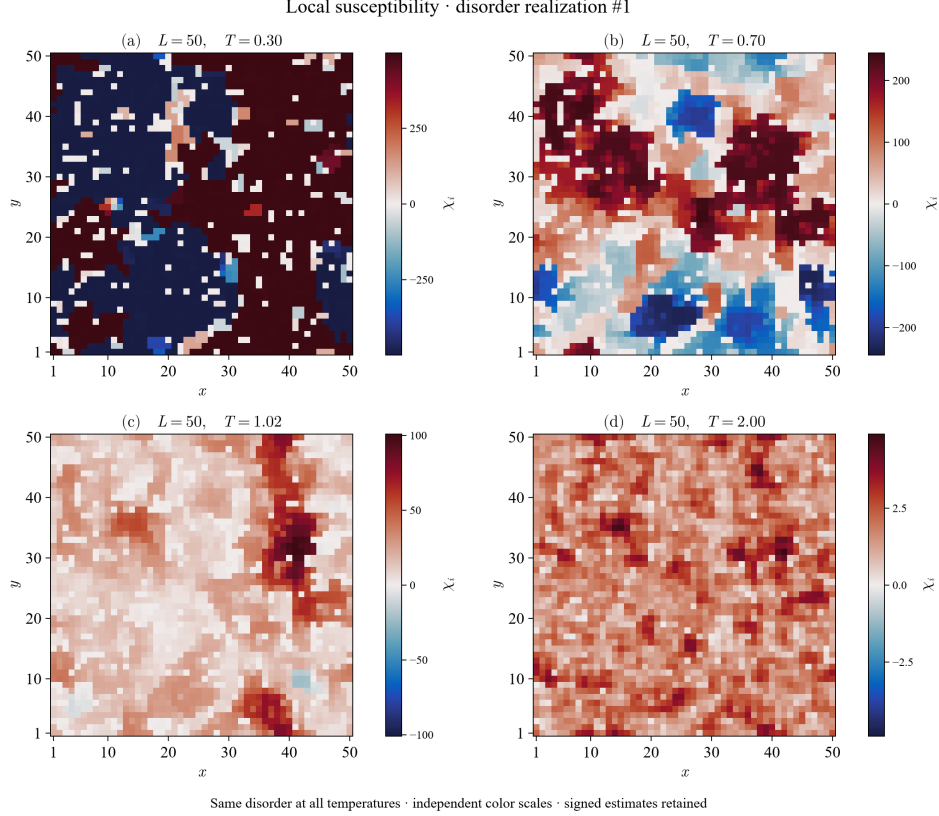
At high temperature, separation between sizes also contains the ordinary extensive factor: asymptotically  $\chi \sim N/T$  for weakly correlated spins. Separation of total-susceptibility curves alone is consequently insufficient evidence for anomalous scaling. At low temperature, heat-bath trajectories may fail to average over relative cluster orientations, biasing  $\langle M^2 \rangle$ . Narrow disorder error bands do not exclude such a problem. The response enhancement must therefore be considered together with the local maps and dynamical measurements below.

### 3.2 Spatial structure of the local susceptibility

The response of an individual spin to a uniform field is

$$\chi_{i,a} = \frac{\langle s_i M \rangle_a}{T} = \frac{1}{T} \sum_j \langle s_i s_j \rangle_a, \quad \sum_i \chi_{i,a} = \chi_a. \quad (4)$$

Figure 3 follows the first  $L = 50$  disorder realization across four temperatures, so the bond geometry remains fixed.



**Figure 3:** Local susceptibility estimates at  $T = 0.30, 0.70, 1.02, 2.00$  for the same bond realization. Each panel uses an independent symmetric linear color scale. Negative estimates are retained; equal color intensity across panels does not mean equal response.

At  $T = 0.30$ , broad patches have nearly uniform estimated response, indicating coherent spin behavior over the measured trajectory. At  $T = 0.70$ , these patches become more structured, while strong spatial variation persists. At  $T = 1.02$ , large positive responses are more concentrated. By  $T = 2.00$ , the scale has fallen from hundreds to values of order unity and finer variations dominate. Residual heterogeneity is expected because local coordination and connectivity remain nonuniform even when thermal correlations are short.

The negative patches require care. Ferromagnetic equilibrium correlations satisfy  $\langle s_i s_j \rangle_a \geq 0$ , implying  $\chi_{i,a} \geq 1/T$ . Negative estimates cannot describe equilibrium antiferromagnetic regions. If disconnected clusters rarely reverse during a finite trajectory, their relative orientations do not average out; cross-cluster terms in  $s_i M$  can remain spuriously negative. This is a natural explanation for the coherent negative patches, although the stored map does not identify every slow mode.

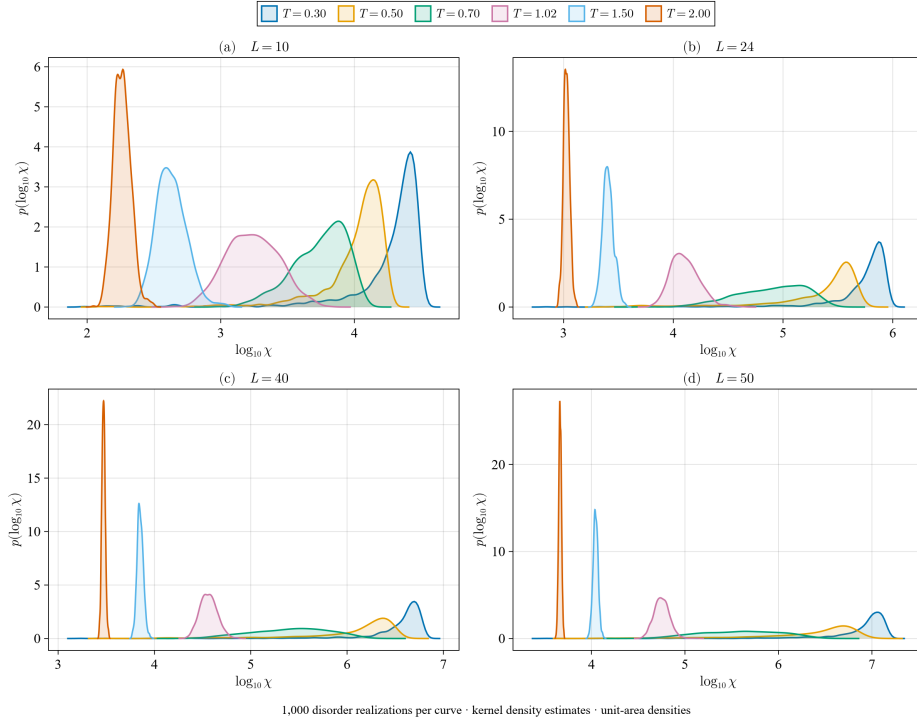
At  $T = 0.30$ , approximately 44.5% of sites have negative estimates and 40.8% fall below minus three times their estimated block SEM. These are not merely small fluctuations within the reported errors. They indicate unresolved sampling or error-estimation problems. Thus the maps reveal spatial structure in measured trajectories, but the low-temperature amplitudes cannot yet be treated as converged equilibrium rare-region responses.

### 3.3 Sample-to-sample susceptibility distributions

Figure 4 shows kernel density estimates of  $x = \log_{10} \chi$  from 1000 realizations at each size and temperature. Their normalization is

$$\int p_x(x) dx = 1, \quad p_\chi(\chi) = \frac{p_x(\log_{10} \chi)}{\chi \ln 10}. \quad (5)$$

The ordinate is a density per unit logarithmic susceptibility, not per unit  $\chi$ .



**Figure 4:** Distributions of total susceptibility at six temperatures for  $L = 10, 24, 40, 50$ , normalized in  $d(\log_{10} \chi)$ . Smooth curves are kernel density estimates; small shoulders require bandwidth and sampling checks before interpretation.

Cooling moves the distributions toward larger susceptibility, consistent with Fig. 2. Their shapes reveal information hidden by the mean. At  $T = 0.70$ , the larger sizes have particularly broad distributions extending across several decades. Samples at identical  $L$  and  $T$  can therefore respond very differently. Variations in cluster sizes and weak connections offer a plausible origin, affecting both correlated magnetization and the time required to explore its orientations.

The width is not monotonic in temperature. The  $T = 0.30$  and  $0.50$  curves concentrate at larger responses and have appreciable tails toward smaller values, whereas the intermediate-temperature distribution is especially broad. This may reflect a crossover between nearly rigid clusters and thermally disrupted correlations. However, slow sampling can also reshape the distributions, and these data do not separate the two contributions.

At  $T = 1.50$  and  $2.00$ , the distributions narrow as  $L$  increases, consistent with averaging over many weakly correlated regions. Their rightward shift is partly the trivial effect of total susceptibility increasing with volume. Replacing  $\chi$  by  $\chi/N$  translates each logarithmic distribution by  $-\log_{10} N$  without changing its width.

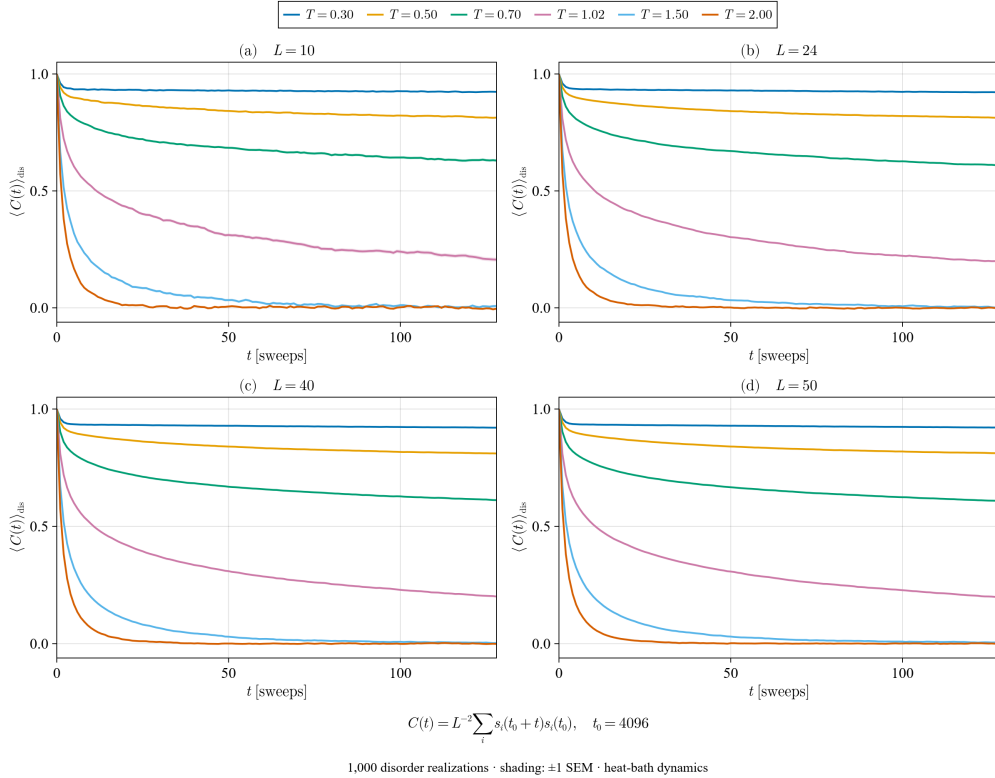
A broad distribution can make the mean unrepresentative of typical samples, but does not establish a power-law tail or a lack of self-averaging. Such conclusions require converged sampling and size dependence of, for example, the relative variance. These densities contain both disorder variation and MC estimation noise; their appearance alone cannot determine a Griffiths exponent.

### 3.4 Autocorrelation under heat-bath dynamics

A reference state is saved at  $t_0 = 4096$  heat-bath sweeps after thermalization. The measured overlap is

$$A_a(t) = \frac{1}{N} \sum_i s_i(t_0 + t) s_i(t_0), \quad 0 \leq t \leq 128. \quad (6)$$

We use  $A(t)$  to distinguish it from heat capacity; the original plot labels it  $C(t)$ . Each sample contributes one fixed-origin trajectory, without time-origin averaging or subtraction of spin means.



**Figure 5:** Disorder-averaged overlap with  $\pm 1$  SEM. One time unit is one random-site heat-bath sweep. The reference state is taken 4096 sweeps after thermalization, with  $A(0) = 1$ . SW updates are excluded from the time axis.

Temperature has a much larger effect than the visible size dependence. At  $L = 50$ ,  $[A(128)]_{\text{dis}} \simeq 0.921, 0.812, 0.609, 0.199$  for  $T = 0.30, 0.50, 0.70, 1.02$ , respectively. At  $T = 1.50$  and  $2.00$ , the overlap is close to zero by the end of the window, with  $T = 2.00$  losing memory substantially faster.

The low-temperature curves show a rapid initial drop followed by much slower decay. A cluster picture suggests that weakly constrained spins relax first, while strongly correlated regions retain memory. Reversing such regions through local updates may require rare intermediate configurations, even though SW updates can flip their overall orientation efficiently. The plotted relaxation therefore characterizes the chosen local heat-bath dynamics, not the autocorrelation time of an SW sampling algorithm.

Similar curves across sizes indicate modest size dependence within this window, rather than size-independent asymptotic relaxation. An apparent plateau over 128 sweeps does not establish frozen order or a nonzero infinite-time limit. For any finite system at positive temperature, heat-bath dynamics can eventually sample both orientations. Longer trajectories are needed to resolve the tail; the linear-axis plots do not distinguish power-law, stretched-exponential, or other decay forms. The SEM combines disorder variation with single-trajectory noise, and adjacent times are correlated.

## 4 Summary

The four figures provide complementary views of the  $p = 0.5$  bond-diluted Ising model. The heat capacity identifies a broad energy-fluctuation crossover near  $T = 1$ , with an approximately size-independent per-site peak. Magnetic susceptibility is more sensitive to large correlated clusters and grows strongly upon cooling. Its distributions show that this enhancement varies substantially between disorder realizations, especially at intermediate temperatures.

The local maps and temporal overlap connect this behavior to spatial heterogeneity and slow heat-bath dynamics. These observations are compatible with a cluster-based picture of the Griffiths region, but at the percolation threshold critical connectivity must also be considered. Neither the heat-capacity maximum nor finite-time memory identifies a finite-temperature ferromagnetic transition.

A central limitation is that low-temperature local-response estimates violate equilibrium positivity. Their magnitude relative to the reported block errors indicates that static sampling is not uniformly converged. Narrow error bands on disorder averages do not remove this concern, and apparent distribution tails may also be affected. The results therefore document finite-size responses and finite-time relaxation, rather than establish a Griffiths singularity or a dynamical exponent.

For quantitative conclusions, static observables should first be checked using efficient cluster sampling and independent runs, while local heat-bath updates are retained for the specified dynamics. Longer observation windows, multiple time origins or independent trajectories, and block-length checks would then resolve slow relaxation more reliably. Separating equilibrium sampling from the chosen stochastic dynamics is essential for identifying robust physical features.